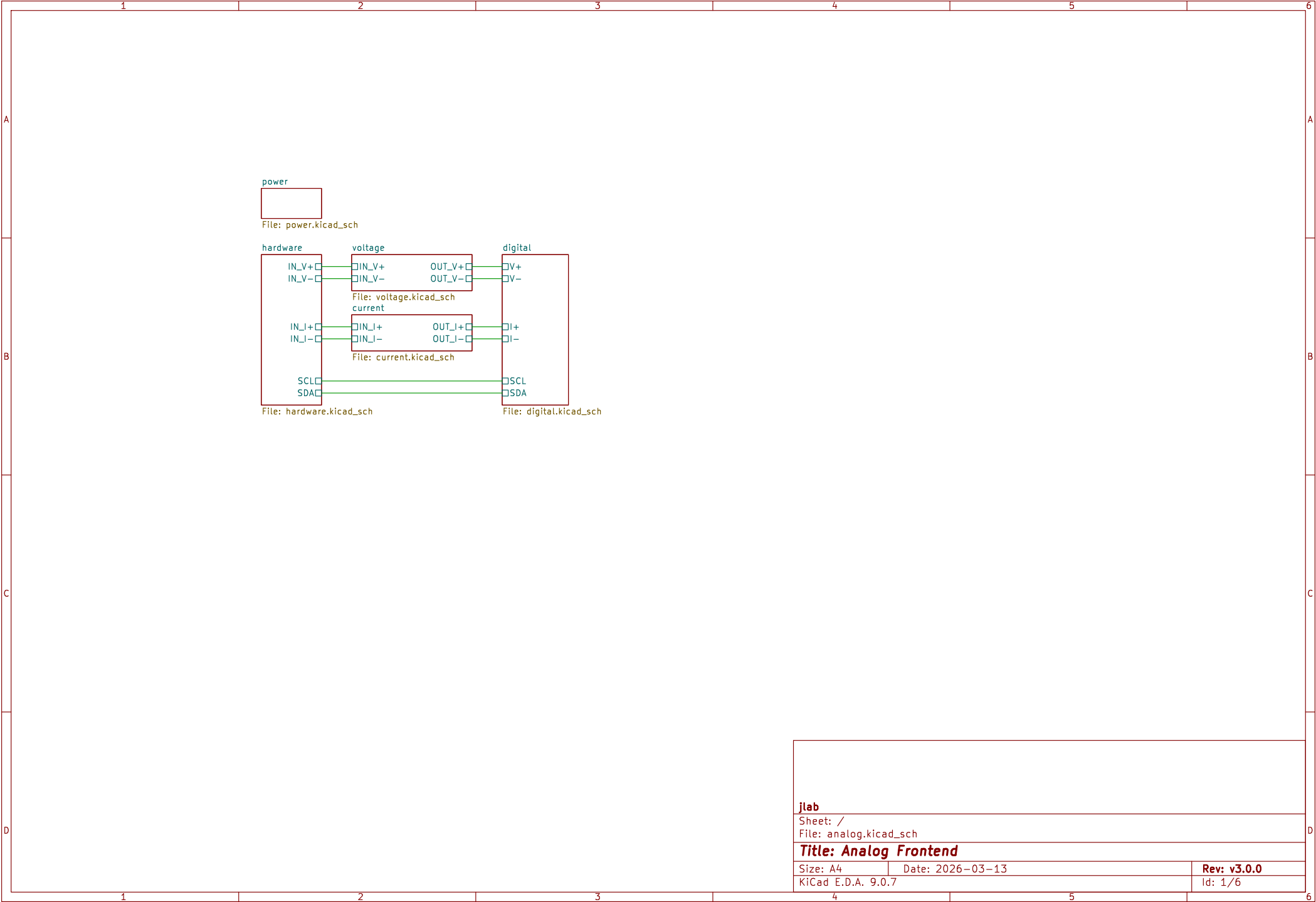




jlalab

Sheet: /

File: analog.kicad_sch

Title: Analog Frontend

Size: A4

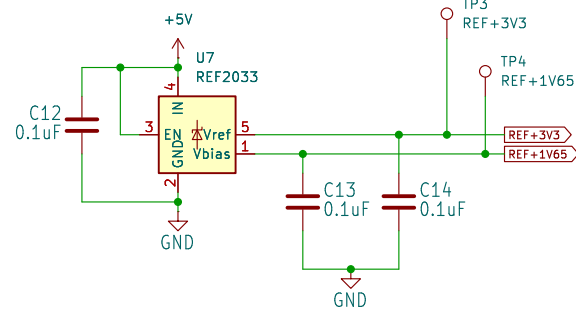
Date: 2026-03-13

KiCad E.D.A. 9.0.7

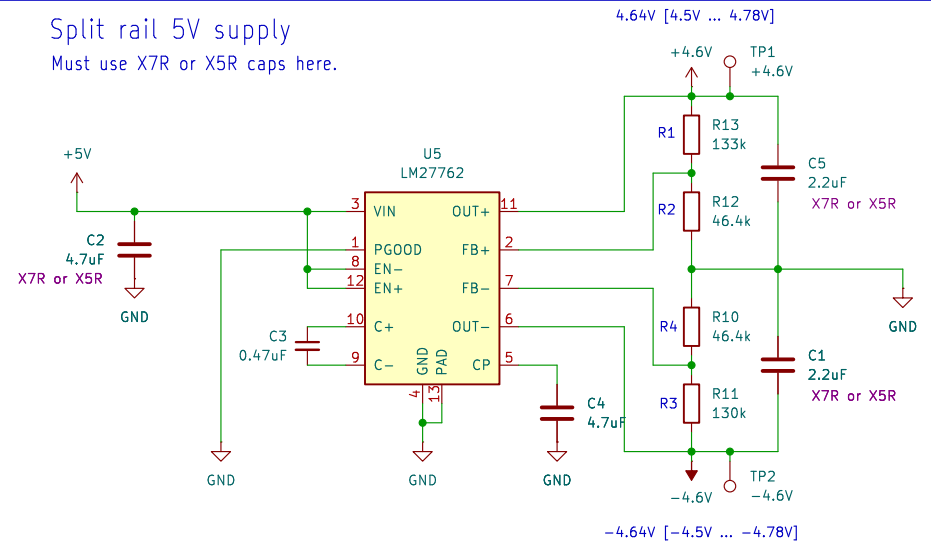
Rev: v3.0.0

Id: 1/6

Place bypass caps close to package.



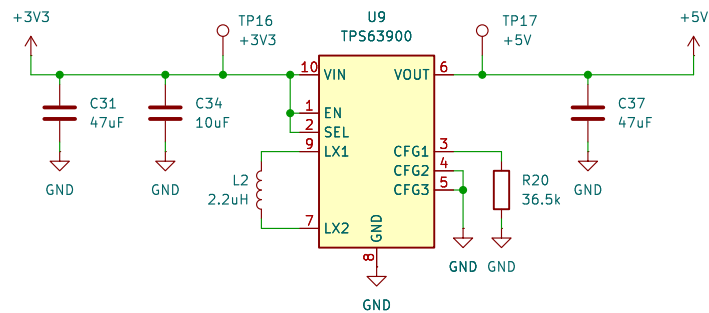
Must use X7R or X5R caps here.



$$\begin{aligned} V_{OUT} &= 1.2 \text{ V} * (R_1 + R_2) / R_1 \\ V_{OUT} &= 1.2 \text{ V} * (133 + 46.4) / 133 \\ V_{OUT} &= 4.64 \text{ V} \end{aligned}$$

$$\begin{aligned} &\text{Need } 0.2 \text{ V overhead on input} \\ V_{OUT} &= -1.22 \text{ V} * (R_3 + R_4) / R_3 \\ V_{OUT} &= -1.22 \text{ V} * (130 + 46.4) / 130 \\ V_{OUT} &= -4.64 \text{ V} \end{aligned}$$

Boost to 5V



Sheet: /power/
File: power.kicad_sch

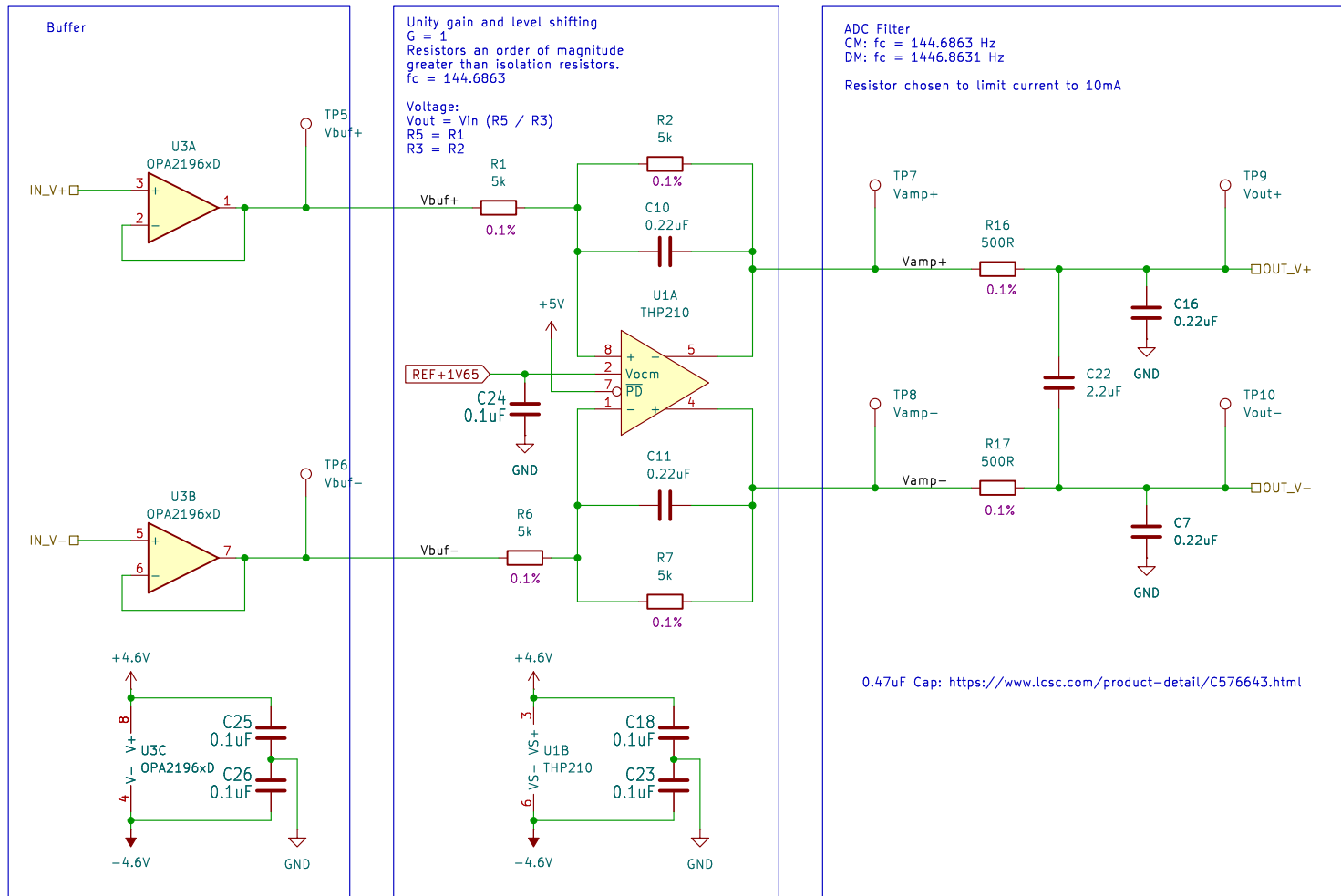
Title: Analog Frontend

Size: A4	Date: 2026-03-13
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KiCad E.D.A. 9.0.7

Rev: v3.0.0

Id: 2/6



jlab

Sheet: /voltage/
File: voltage.kicad_sch

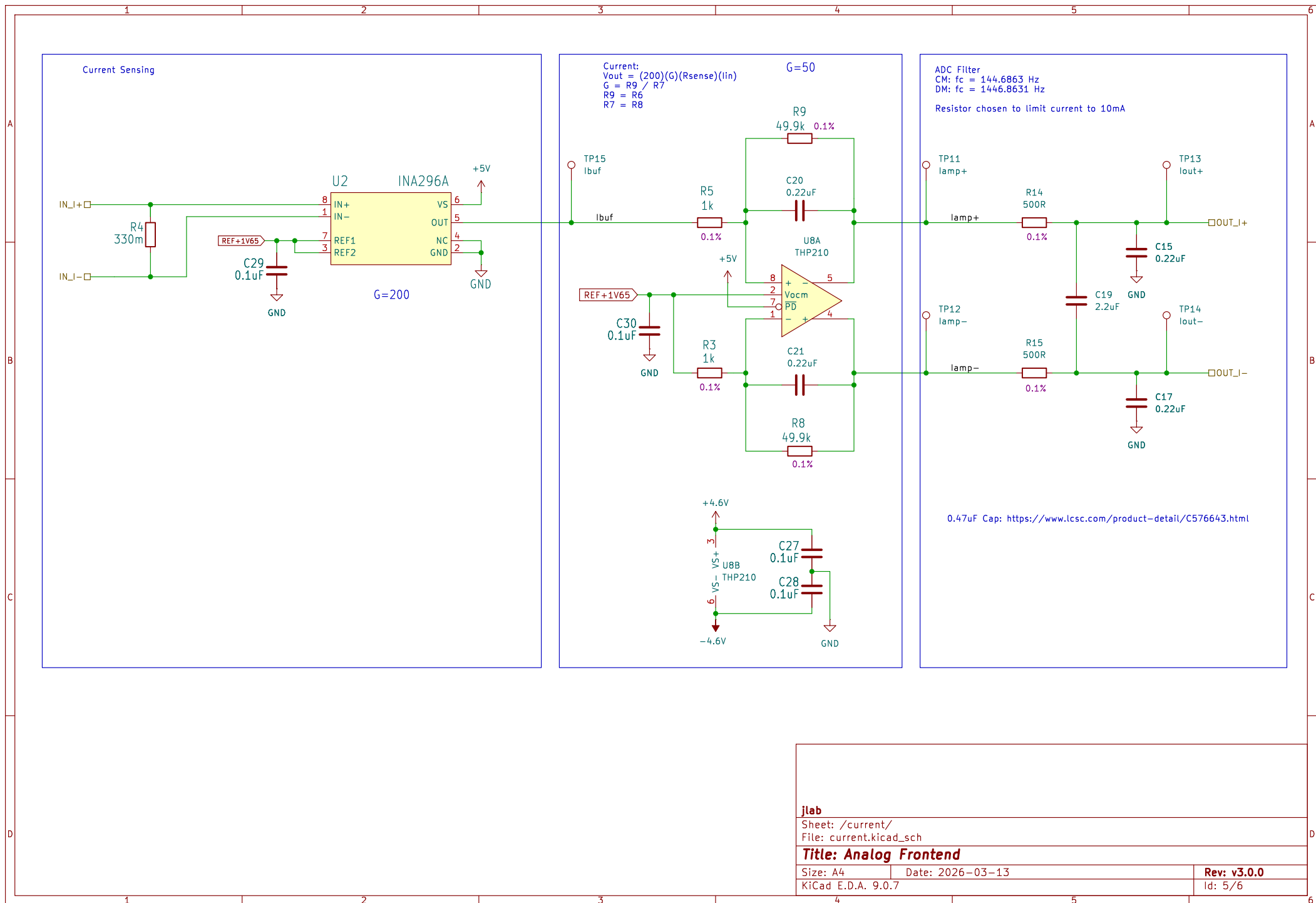
Title: Analog Frontend

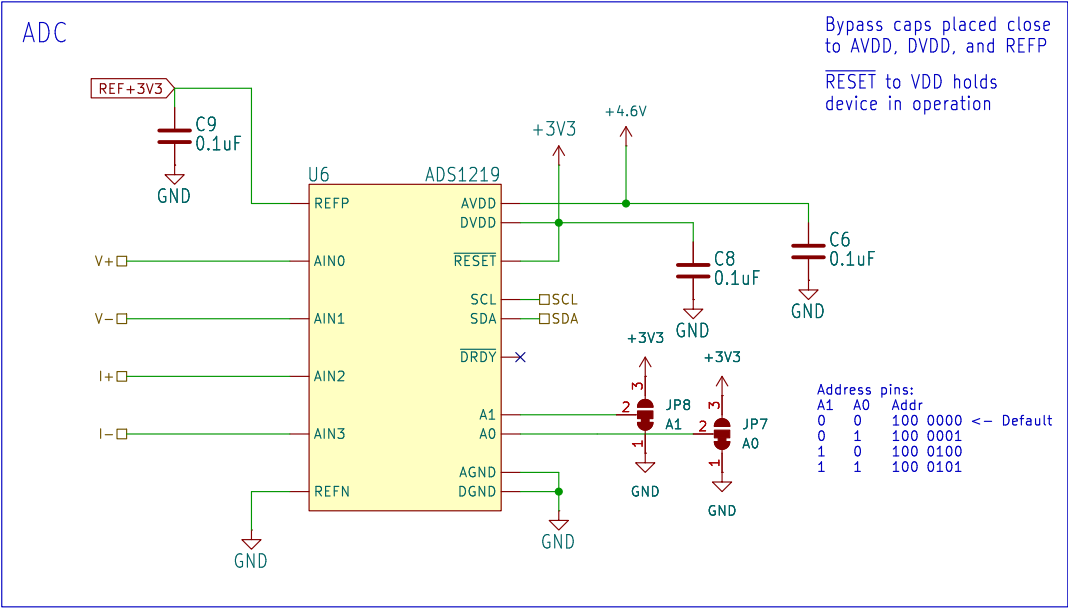
Size: A4 Date: 2026-03-13

KiCad E.D.A. 9.0.7

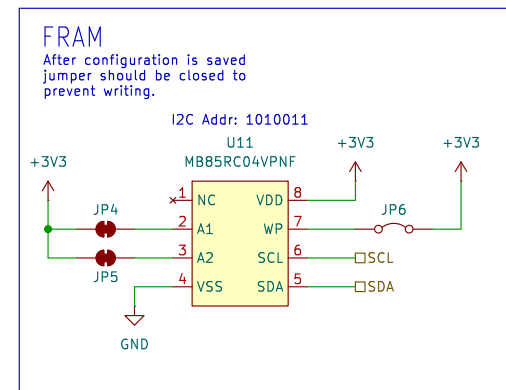
Rev: v3.0.0

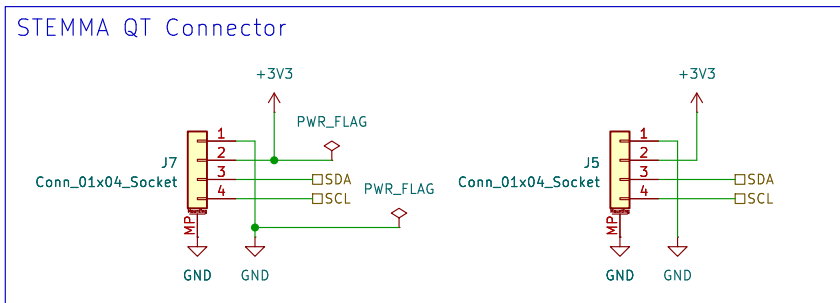
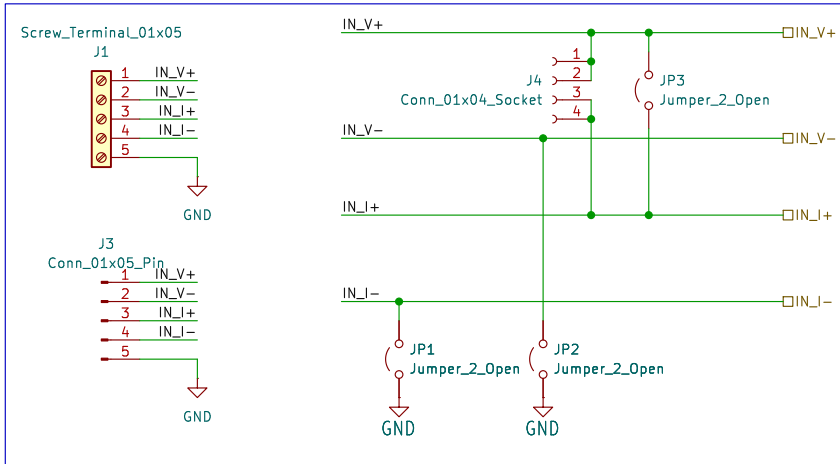
Id: 4/6





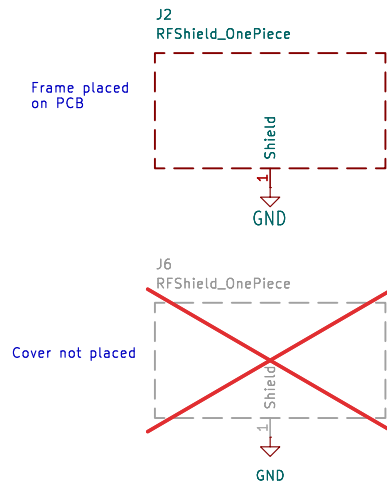
TODO: Check conflicting addresses with other parts



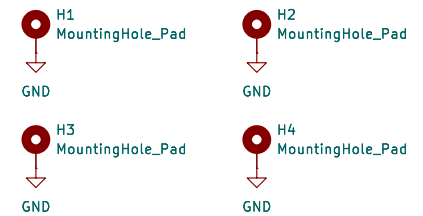


RF Shield

Covers analog components in sheet 1 and reference voltage source.



Mounting Holes



jlab

Sheet: /hardware/
File: hardware.kicad_sch

Title: Analog Frontend

Size: A4 Date: 2026-03-13

KiCad E.D.A. 9.0.7

Rev: v3.0.0

Id: 7/6